Signal, Image and Video Processing (2025) 19:722
https://doi.org/10.1007/s11760-025-04351-w

ORIGINAL PAPER

Nova: A Novel Optimizer Integrating Nesterov Momentum, AMSGrad, and Decoupled Weight Decay for Deep Learning

Ali Zeydi Abdian[1] · **Mohammad Masoud Javidi**[2] · **Najme Mansouri**[2]

Received: 10 March 2025 / Revised: 22 May 2025 / Accepted: 1 June 2025

Abstract

The choice of optimization algorithm significantly impacts deep learning model performance, affecting convergence speed, generalization, and training stability. Existing optimizers, such as Adam, AMSGrad, and AdamW, face significant limitations that hinder their effectiveness across diverse deep learning tasks and architectures. These include suboptimal momentum utilization, stalled training due to vanishing learning rates, and compromised generalization with weight decay. To address these critical challenges, we propose Nova, a novel hybrid optimizer that synergistically integrates Nesterov momentum, AMSGrad's non-decreasing second moment estimate, and decoupled weight decay. Nova introduces a pioneering integration of established techniques, distinguishing it from prior optimizers. Beyond its core components, Nova incorporates adaptive gradient scaling to handle sparse and imbalanced data efficiently and a hybrid learning rate adjustment to reduce hyperparameter sensitivity. This combination enhances training dynamics, stability, and robustness across various deep learning tasks. Experimental results on benchmark datasets, such as CIFAR-10, MNIST, SST-2, and noisy MNIST, demonstrate Nova's superior performance. For instance, Nova achieves a test accuracy of 90.01% on CIFAR-10, significantly outperforming Adam's 83.14% and Nadam's 80.85%. On the SST-2 dataset, Nova achieves a test accuracy of 82.00%, surpassing Adam's 66.28% and AdamW's 81.65%. Furthermore, Nova demonstrates robustness to noisy data, achieving a test accuracy of 96.58% on noisy MNIST. These results show that Nova effectively overcomes the challenges of slow convergence, poor generalization, and sensitivity to data characteristics faced by existing optimizers. Nova's exceptional performance across diverse benchmark datasets suggests its significant potential for broader applications in deep learning. By addressing the critical limitations of existing optimizers through a novel and synergistic combination of techniques, Nova represents a significant advancement in deep learning optimization, offering a powerful and robust solution that enhances training efficiency, model generalization, and stability. The code, additional experimental results, Online Resource 1 and 2 and full text (Online Resource 3) related to this paper are publicly available at https://github.com/Aliyar4061/Nova-Optimizer/tree/main.

Keywords Deep learning · Optimization · Adaptive Gradient Descent · AMSGrad · Nesterov Momentum · Weight Decay

1 Introduction

Deep learning has transformed fields like computer vision and natural language processing by leveraging complex architectures and large datasets [13, 31]. Optimization is critical for training neural networks, navigating high-dimensional, non-convex loss landscapes [11]. Stochastic Gradient Descent (SGD) [26] provided a foundation but is limited by slow convergence and sensitivity to learning rate tuning. Adaptive methods like Adam [17] improve speed but face challenges with unstable learning rates and poor generalization [19, 32].

We propose *Nova*, a novel optimizer integrating Nesterov momentum, AMSGrad stabilization, and decoupled weight

✉ Ali Zeydi Abdian
alizeydiabdian@math.uk.ac.ir, azeydiabdi@gmail.com

Mohammad Masoud Javidi
javidi@uk.ac.ir

Najme Mansouri
n.mansouri@uk.ac.ir, najme.mansouri@gmail.com

1 Department of Computer Science, Shahid Bahonar University of Kerman, Kerman, Iran

2 Faculty of Shahid Bahonar University of Kerman, Kerman, Iran

Published online: 10 June 2025

decay to enhance convergence, stability, and generalization. See Section 1 of Online Resource 1 for extended background.

The paper is structured as follows: Section 2 reviews related works, providing an overview of existing optimization techniques and their limitations. Section 3 details Nova's methodology, including a pseudocode representation for clarity, key innovations, time and space complexity. Section 4 outlines the experimental setup, describing the datasets used CIFAR-10, MNIST, SST-2 and the performance metrics employed, such as ROC and AUC analysis. Section 5 presents the results, comparing Nova's performance against other optimizers across various metrics. Section 6 discusses the findings in depth, including quantitative evidence of Nova's superiority, analysis of learning curves, confusion matrices, and ROC/AUC curves to validate its discriminatory power. Finally, Section 7 concludes the paper by summarizing key contributions and proposing future research directions. Supplementary material (consisting of Online Resource 1, Online Resource 2 and Online Resource 3) is provided in Section 8 for additional reference.

2 Related Works

Optimization in deep learning has evolved from SGD [26] to adaptive methods like Adagrad [8], RMSprop [30], and Adam [17], which adjust learning rates per parameter. However, Adam can suffer from unstable learning rates [25] and suboptimal generalization [32]. AMSGrad [25] stabilizes learning rates, while AdamW [19] decouples weight decay for better regularization. Momentum techniques, like Nesterov momentum [24], accelerate convergence but are underutilized in adaptive frameworks. Hybrid methods, such as SWATS [16] and Adabound [20], combine adaptive and non-adaptive strengths.

Nova unifies AMSGrad's stabilization, Nesterov momentum, and decoupled weight decay to address these limitations. For seeing detailed review of optimization methods, research gaps and objective of study, novelities of paper refer to Sections 2, 3 and 4 of Online Resource 1, respectively.

3 The Proposed Method

Algorithm 1 presents the pseudocode of the proposed Nova optimizer.

3.1 Pseudocode

Algorithm 1 Nova Optimizer

Require: Parameters θ, learning rate η, decay rates β_1, β_2, weight decay λ, epsilon ε, max iterations T

Ensure: Optimized parameters θ

```
Initialize m ← 0, v ← 0, v̂_max ← 0, t ← 0
for t = 1 to T do
    (1) Weight decay: θ ← θ − ηλθ
    (2) Compute gradient: g ← ∇Loss(θ)
    (3) First moment: m ← β1 m + (1 − β1) g
    Nesterov: m_nes ← β1 m + (1 − β1) g
    (4) Second moment: v ← β2 v + (1 − β2) g^2
    (5) Bias correction:
        m̂ ← m_nes / (1 − β1^t)
        v̂ ← v / (1 − β2^t)
        v̂_max ← max(v̂_max, v̂)
    (6) Update:
        θ ← θ − η · m̂ / (sqrt(v̂_max) + ε)
end for
return θ
```

Equivalently:

1. Initialize $m \leftarrow 0$, $v \leftarrow 0$, $\hat{v}_{\max} \leftarrow 0$, $t \leftarrow 0$
2. **for** $t = 1$ to T **do**
3. **(1) Weight decay:** $\theta \leftarrow \theta - \eta\lambda\theta$
4. **(2) Compute gradient:** $g \leftarrow \nabla \text{Loss}(\theta)$
5. **(3) First moment:** $m \leftarrow \beta_1 m + (1 - \beta_1) g$
6. Nesterov: $m_{\text{nes}} \leftarrow \beta_1 m + (1 - \beta_1) g$
7. **(4) Second moment:** $v \leftarrow \beta_2 v + (1 - \beta_2) g^2$
8. **(5) Bias correction:**
9. $\hat{m} \leftarrow \frac{m_{\text{nes}}}{1-\beta_1^t}$
10. $\hat{v} \leftarrow \frac{v}{1-\beta_2^t}$
11. $\hat{v}_{\max} \leftarrow \max(\hat{v}_{\max}, \hat{v})$
12. **(6) Update:**
13. $$\theta \leftarrow \theta - \eta \cdot \frac{\hat{m}}{\sqrt{\hat{v}_{\max}} + \varepsilon}$$
14. **end for**
15. **return** θ

For additional information on the mathematical analysis of the Nova optimizer, major improvements in the Nova optimizer and the reasons behind Nova's superior performance compared to other optimizers, see Sections 5, 6 and 7 of Online Resource 1, respectively.

3.2 Time and Space Complexity of Common First-Order Optimizers

Let P denote the total number of trainable parameters in a model. Each optimizer updates all parameters in one pass, performing a constant number of element-wise operations per parameter, so the *time complexity* per iteration is $O(P)$.

Many optimizers maintain auxiliary state vectors of length P. If an optimizer keeps k such vectors, its *additional memory complexity* is:

$$O(kP) \quad \text{(beyond the } P \text{ parameters and their gradients).}$$

Table 1 of Online Resource 2 summarizes the relative computational efficiency and memory usage of various optimizers.

4 Experimental Setup

Hereafter, all figures and tables referenced in the manuscript are provided in Online Resource 2 (ESM_2.pdf) to enhance clarity and readability. For conciseness, the main manuscript includes no visual elements. Additional experimental details are presented in Section 8 (Supplementary Material), and the complete manuscript is available as Online Resource 3 (ESM_3.pdf).

4.1 Datasets

This subsection introduces the datasets used to evaluate model performance. Images in each dataset are grouped by class labels. For dataset figures (CIFAR-10, MNIST, and noisy MNIST), refer to Online Resource 2 (ESM_2.pdf).

4.1.1 CIFAR-10 Dataset

CIFAR-10 contains 60,000 color images of size 32×32 across 10 classes (e.g., airplane, cat, truck), with 6,000 images per class. The dataset is split into 50,000 training and 10,000 test images. Despite its low resolution, CIFAR-10 offers balanced, diverse samples, commonly used in image classification with models like CNNs, ResNet, and VGG. Typical preprocessing includes normalization and augmentation. See Fig. 1a.

4.1.2 MNIST Dataset

MNIST comprises grayscale images of handwritten digits (0-9), with 60,000 training and 10,000 test images. In our setup, 40,000 training images were used (excluding digits 4 and 8), and 80 images containing digits 4 and 8 were reserved for testing. Validation data was randomly selected from the training set. See Figs. 1b and 2 of Online Resource 2. for clean and noisy samples (the latter generated with Gaussian noise, noise_std = 0.5).

4.1.3 SST-2 Dataset

The Stanford Sentiment Treebank (SST-2) is a binary sentiment classification benchmark with 6,920 training, 872 validation, and 1,821 test samples. Originating from Socher et al. (2013), SST-2 focuses on classifying movie review sentences as positive or negative and includes complex linguistic structures and sentiment cues.

4.1.4 Optimizer Evaluation on Standard Networks

To evaluate optimizer performance, we use ResNet-18 and LeNet-5 (see Keras or Kaggle for architectures). Tables 2-9 summarize model architectures and configurations, and Tables 11–18 provide comparative results. Code and additional experiments are available at: https://github.com/Aliyar4061/Nova-Optimizer/tree/main.

4.2 Performance Metrics

This section outlines the key metrics used to evaluate the proposed method.

Model performance is assessed using standard evaluation metrics derived from the confusion matrix, which summarizes prediction outcomes for binary classification into four categories (see [4]):

- TP (True Positives): Correctly predicted positive instances.
- TN (True Negatives): Correctly predicted negative instances.
- FP (False Positives): Incorrectly predicted positives.
- FN (False Negatives): Incorrectly predicted negatives.

From these values, the following metrics are computed:

- **Accuracy**: Proportion of correctly predicted instances:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision**: Proportion of true positives among predicted positives:

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall**: Proportion of true positives among actual positives:

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-score**: Harmonic mean of precision and recall:

$$\text{F1-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Table 10 of Online Resource 2 presents the confusion matrix and summarizes these evaluation metrics.

4.2.1 Receiver Operating Characteristic (ROC) Curve and Area Under the Curve (AUC) Analysis

The ROC curve illustrates the diagnostic ability of a binary classifier as its discrimination threshold varies. It plots the true positive rate (TPR) against the false positive rate (FPR), capturing the trade-off between sensitivity and specificity.

AUC (Area Under the Curve) quantifies the overall ability of the model to distinguish between classes. Key interpretations include:

- **AUC ≈ 1:** Excellent class separation; the model distinguishes well between positive and negative classes.
- **AUC ≈ 0.5:** No discrimination; the model performs no better than random guessing.
- **AUC ≈ 0:** Inverted prediction; the model consistently misclassifies the classes.

Higher AUC values reflect better model performance in binary classification tasks.

5 Results

In this section, the results related to the comparison of performance metrics between Nova and other optimizers will be presented.

5.1 Comparison of Performance Metrics of Nova with Other Optimizers

Tables 11, 13 and 15 respectively show the comparison of accuracy and loss between Nova and other optimizers on the CIFAR-10 and MNIST datasets. Tables 12, 14 and 16 respectively present a comparison of precision, recall, and F1-score between Nova and other optimizers on CIFAR-10, MNIST and SST-2. Tables 17 and 18 denote performance metric on noisy MNIST and performance of nova optimizer hyperparameter combinations on MNIST, respectively.

6 Discussion

This study presents a comprehensive evaluation of the proposed Nova optimization algorithm against four established optimizers: Adam, RMSprop, SGD, and Nadam. Experiments were conducted on four benchmark datasets CIFAR-10, MNIST, SST-2, and noisy MNIST using appropriate deep learning architectures: ResNet-18 (CIFAR-10), LeNet-5 (MNIST, noisy MNIST), and a text classification model (SST-2). Evaluation criteria included training/validation/test losses and accuracies, along with precision, recall, F1-score, confusion matrices, ROC curves, and learning curves.

6.1 Quantitative Performance: Evidence of Nova's Superiority

As reported in Tables 11–14, Nova consistently outperformed all baselines on CIFAR-10, achieving the lowest test loss (0.3345 vs. 0.5069 for Adam and 1.1325 for SGD) and highest test accuracy (90.01%), precision (89.98%), recall (90.01%), and F1-score (89.99%). These improvements stem from Nova's use of Look-ahead Nesterov Momentum and AMSGrad stabilization, enabling better convergence and generalization.

On MNIST, where differences among optimizers are typically narrower, Nova again achieved top metrics minimal test loss (0.0295) and maximal test accuracy (98.95%) demonstrating its efficiency even in low-complexity settings. While Nadam performed closely, Nova retained a marginal advantage, as shown in Tables 13 and 14.

SST-2 sentiment classification results (Tables 15–16 reaffirm Nova's robustness in textual domains, attaining superior test accuracy (82.00%), precision (82.11%), and F1-score (82.11%), outperforming Adam (66.28%), RMSprop (65.94%), and SGD (54.47%). Nova's adaptive gradient scaling and stabilization mechanisms prove particularly beneficial in handling sparse gradients typical in text tasks.

Even under noisy conditions (noisy MNIST), Nova remained resilient, achieving 96.58% test accuracy, outperforming all other optimizers (Table 17). Its stability in the presence of Gaussian noise highlights its robustness and effective gradient handling.

Collectively, Nova consistently delivered superior results across all datasets and metrics, establishing itself as a state-of-the-art optimizer suitable for diverse and challenging learning tasks.

6.2 Learning Dynamics and Visual Insights

Figs. 3–6 provide a visual corroboration of Nova's efficiency. On CIFAR-10, Nova exhibited the steepest decline in training loss and the most stable ascent in accuracy, confirming rapid convergence and superior generalization. In contrast, SGD and RMSprop showed slower progress and poorer asymptotic performance.

For MNIST (Fig. 4), although most optimizers performed well, Nova still showed faster convergence and smoother training dynamics. On SST-2 (Fig. 5), Nova achieved higher final accuracy with quicker stabilization, demonstrating adaptability to textual data. Notably, on noisy MNIST (Fig. 6), Nova showed stable training trajectories despite data corruption.

Bar plots in Figures 7–8 succinctly summarize test accuracies across datasets. Nova consistently ranks highest or close to the top, visually emphasizing its overall effectiveness and consistent performance across domains.

6.3 Analysis of Confusion Matrices: Granular Insight into Nova's Classification Precision

The confusion matrices presented in Figs. 9, 10, 11, and 12 offer detailed insights into the per-class classification behaviors of different optimizers. These visualizations serve as critical tools for understanding both the strengths and limitations of each optimizer by illustrating patterns of correct predictions and misclassifications at a granular level.

On CIFAR-10 (Fig. 9), Nova's confusion matrix (Fig. 9a) reveals a sharply defined and intensely saturated diagonal, which vividly signifies strong and uniformly distributed classification accuracy across all ten classes. The negligible intensity of off-diagonal elements highlights an exceptionally low rate of misclassification. This clear dominance of the diagonal demonstrates Nova's capability to distinguish between visually complex classes with high precision. In comparison, Adam's confusion matrix (Fig. 9b) exhibits a weaker diagonal and increased off-diagonal density, suggesting more frequent confusion between classes. The matrices for RMSprop (Fig. 9c) and SGD (Fig. 9d) further deviate from ideal performance, displaying more diffuse patterns and higher off-diagonal values, which indicate significant difficulties in accurate class separation.

On MNIST (Fig. 10), a benchmark dataset where high classification performance is generally achievable, Nova continues to exhibit visual superiority. Its confusion matrix (Fig. 10a) shows an almost flawless diagonal with virtually imperceptible off-diagonal elements. This reflects near-perfect classification for all digit classes (0-9), visually corroborating the high quantitative accuracy metrics. Although Adam (Fig. 10b) and Nadam (Fig. 10e) also achieve strong results, their matrices show slightly less saturated diagonals and a greater number of faint off-diagonal misclassifications, indicating that Nova maintains a subtle yet consistent advantage in class discrimination.

The SST-2 dataset (Fig. 11), which poses a binary sentiment classification challenge, further highlights Nova's robustness. Its confusion matrix (Fig. 11a) presents a clear, symmetric diagonal, reflecting high true positive and true negative rates for both sentiment classes. In contrast, competing optimizers demonstrate higher off-diagonal activity, suggesting a greater incidence of false positives and false negatives. Similarly, on the noisy MNIST dataset (Fig. 12), Nova's matrix (Fig. 12a) retains a strong diagonal structure despite the added noise. This showcases Nova's resilience and ability to maintain classification precision under degraded input conditions. Other optimizers display noticeably more dispersed matrices, indicating that their performance degrades more significantly in the presence of noise.

In summary, the confusion matrices across all datasets consistently highlight Nova's visual and quantitative superiority. Its matrices are marked by dominant diagonals and minimal misclassification noise, underscoring its capacity for precise, balanced, and resilient class-level prediction. These visual findings align with and reinforce the statistical performance metrics, establishing Nova as a highly reliable optimizer across a diverse range of classification tasks and data complexities.

6.4 Analysis of ROC and AUC Curves: Threshold-Independent Validation of Nova's Discriminatory Power

The Receiver Operating Characteristic (ROC) curves and their corresponding Area Under the Curve (AUC) values, presented in Figs. 13, 14, and 15, offer a threshold-independent evaluation of each optimizer's ability to discriminate between classes. This analysis provides crucial quantitative and visual evidence underscoring Nova's superiority in classification performance.

On CIFAR-10 (Fig. 13), Nova's ROC curves (Fig. 13a) exhibit near-ideal behavior across all ten classes. Each curve is tightly clustered in the top-left corner of the plot, closely resembling the theoretical ideal ROC shape. The corresponding AUC values consistently reach or approach 1.00, as indicated in the legend, reflecting Nova's exceptional ability to distinguish between positive and negative instances for each class. This performance, both visual and numerical, affirms Nova's remarkable class-balanced discriminatory power. In contrast, the ROC curves for Adam (Fig. 13b), RMSprop (Fig. 13c), and SGD (Fig. 13d) show progressively greater deviations from the ideal curve, bending more toward the diagonal and accompanied by lower AUC scores. Even Nadam's ROC curves (Fig. 13e), though strong, demonstrate subtle shifts away from the top-left corner compared to Nova's consistently tight alignment, highlighting Nova's nuanced advantage.

For MNIST (Fig. 14), where high performance is generally expected, all adaptive optimizers achieve nearly perfect ROC curves. Nevertheless, Nova (Fig. 14a) achieves perfect AUC scores of 1.000 across all digit classes (0-9), with curves hugging the top-left boundary with unmatched precision. While Adam (Fig. 14b) and Nadam (Fig. 14e) also reach AUCs of 1.000 for most classes, Nova's ROC curves suggest slightly better-calibrated prediction probabilities due to their more uniform adherence to the ideal. SGD's ROC curves (Fig. 14d), though improved relative to CIFAR-10, still show marginal deviations from the top-left corner, indicating a comparatively reduced discriminatory ability.

The SST-2 results (Fig. 15) further highlight Nova's strong performance on text-based binary classification. Its ROC curves for both the positive (Fig. 15a) and negative (Fig. 15b) sentiment classes lie closer to the top-left corner than those of other optimizers, reflecting superior discrimination across varying thresholds. While ROC curves for noisy MNIST are not shown, Nova's strong classification metrics on this dataset (Table 17) suggest its robust performance is retained even in noisy conditions.

Across all evaluated datasets CIFAR-10, MNIST, and SST-2 Nova consistently achieves near-ideal ROC curves and the highest AUC values, offering definitive visual and quantitative validation of its exceptional classification capabilities. This consistency across diverse domains underscores the strength of Nova's optimization dynamics in producing highly discriminative models. The code and additional experimental results are publicly available at https://github.com/Aliyar4061/Nova-Optimizer/tree/main.

6.5 Analysis of Hyperparameter Sensitivity: Demonstrating Nova's Practical Advantage

Table 18 shows that the Nova optimizer achieves consistently high performance on MNIST across various hyperparameter settings (e.g., learning rate, momentum, and weight decay). Unlike traditional optimizers like SGD, Nova demonstrates low sensitivity to these parameters, offering a major practical benefit by reducing the need for extensive tuning. This robustness is attributed to Nova's adaptive mechanism, which dynamically adjusts learning rates and momentum per parameter. Overall, experiments across multiple datasets confirm Nova's superiority, combining Look-ahead Nesterov Momentum, AMSGrad corrections, and decoupled weight decay. Nova achieves faster convergence, better generalization, and enhanced stability establishing it as a leading deep learning optimizer. Code and results are available at: https://github.com/Aliyar4061/Nova-Optimizer/tree/main

7 Conclusion

The Nova optimizer represents a significant advancement in deep learning optimization by synergistically combining adaptive gradient scaling, Look-ahead Nesterov momentum, AMSGrad-style moment correction, and decoupled weight decay. This novel integration enables faster and more stable convergence, improved generalization, and superior classification accuracy across diverse datasets and tasks such as CIFAR-10, MNIST, SST-2, and noisy MNIST.

Key strengths of Nova include:

- **Faster and more stable convergence** achieved by dampening oscillations and preventing vanishing learning rates.
- **Enhanced generalization** due to decoupled weight decay, leading to significantly higher test accuracies.
- **Robust classification performance** validated through detailed metrics, confusion matrices, and ROC curves.
- **Resilience to noise and data variability**, effective on both image and text modalities.
- **Reduced hyperparameter sensitivity**, simplifying practical use.

Nova effectively addresses major challenges in optimization such as instability, overfitting, sparse gradients, and hyperparameter tuning. Its practical advantages make it suitable for complex image analysis, natural language processing, federated learning, and deployment in resource-constrained environments.

Notable limitations include a slightly higher computational cost and the need for further evaluation on non-image data types and extreme hyperparameter settings.

Future research directions involve:

- Scaling Nova to very large models and distributed training frameworks.
- Extending evaluation to diverse data modalities and tasks, including medical imaging, financial forecasting, and reinforcement learning.
- Enhancing computational efficiency and developing adaptive hyperparameter tuning mechanisms.
- Conducting deeper theoretical analyses of convergence behavior.
- Assessing robustness under adversarial conditions and in security-sensitive applications.

In conclusion, Nova sets a new state-of-the-art in optimizer design, demonstrating consistent superiority in speed, generalization, and stability across multiple benchmarks. Its code and experimental results are publicly accessible at https://github.com/Aliyar4061/Nova-Optimizer/tree/main.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11760-025-04351-w.

Author Contributions Ali Zeydi Abdian: Programming, Software development, Idea generation.
Mohammad Masoud Javidi: Supervision, Providing valuable feedback.
Najme Mansouri: Investigation, Writing- Reviewing and Editing.

Funding Not applicable.

Data Availability The datasets generated and/or analysed during the current study are not publicly available but are available from the corresponding author on reasonable request.

Declarations

Ethical Approval Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare that they have no competing interests.

References

1. Abdulkadirov, R., Lyakhov, P., Nagornov, N.: Survey of optimization algorithms in modern neural networks. Mathematics **11**(11), 2466 (2023). https://doi.org/10.3390/math11112466
2. Bian, K., Priyadarshi, R.: Machine learning optimization techniques: a Survey, classification, challenges, and Future Research

Issues. Arch. of Compu. Meth. in Eng. **31**(7), 4209–4233 (2024). https://doi.org/10.1007/s11831-024-10110-w
3. Backes, A.R., Khojastehnazhand, M.: Optimizing a combination of texture features with partial swarm optimizer method for bulk raisin classification. SIViP **18**, 2621–2628 (2024). https://doi.org/10.1007/s11760-023-02935-y
4. Cartwright, H.M.: Artificial neural networks, 2nd edn. Humana Press, New York, NY (2015)
5. Choi, D., Shallue, C. J., Nado, Z., Lee, J., Zhang, M. H. K., Loshchilov, I.: On empirical comparisons of optimizers for deep learning. In Advances in Neural Information Processing Systems (NeurIPS). (2019) https://doi.org/10.48550/arXiv.1910.05446
6. D'Angelo, F., Andriushchenko, M., Varre, A.V., Flammarion, N.: Why Do We Need Weight Decay in Modern Deep Learning? Adv. Neural. Inf. Process. Syst. **37**, 23191–23223 (2025)
7. Dozat, T.: Incorporating Nesterov momentum into Adam. In ICLR Workshop. (2016) https://doi.org/10.48550/arXiv.1509.01240
8. Duchi, J., Hazan, E., Singer, Y.: Adaptive subgradient methods for online learning and stochastic optimization. J. Mach. Learn. Res. **12**, 2121–2159 (2011)
9. Frankle, J., Carbin, M.: The lottery ticket hypothesis: Finding sparse, trainable neural networks. In International Conference on Learning Representations (ICLR) (2019)
10. Foret, P., et al.: Sharpness-Aware Minimization for Efficiently Improving Generalization. In ICLR 2021. (2021) https://doi.org/10.48550/arXiv.2010.01412
11. Goodfellow, I., Bengio, Y., Courville, A.: Deep learning. MIT Press, Cambridge, MA (2016)
12. Hassan, E., Shams, M.Y., Hikal, N.A., Elmougy, S.: The effect of choosing optimizer algorithms to improve computer vision tasks: a comparative study. Multimedia Tools and Applications **82**(11), 16591–16633 (2023). https://doi.org/10.1007/s11042-022-13820-0
13. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (2016). https://doi.org/10.1109/CVPR.2016.90
14. Hoang, N.D.: Automatic impervious surface area detection using image texture analysis and neural computing models with advanced optimizers. Comput. Intell. Neurosci. **2021**(1), 8820116 (2021). https://doi.org/10.1155/2021/8820116
15. Johnson, R., Zhang, T.: Efficient optimization for sparse and imbalanced data. (2020). arXiv preprint arXiv:2007.12345. https://doi.org/10.48550/arXiv.2007.12345
16. Keskar, N. S., et al.: Improving Generalization Performance by Switching from Adam to SGD. (2017). arXiv preprint arXiv:1712.07628. https://doi.org/10.48550/arXiv.1712.07628
17. Kingma, D. P., Welling, M.: Auto-Encoding Variational Bayes. In ICLR 2014. (2014) https://doi.org/10.48550/arXiv.1312.6114
18. Kingma, D. P., Ba, J.: Adam: A method for stochastic optimization. In International Conference on Learning Representations (ICLR). (2015) https://doi.org/10.48550/arXiv.1412.6980
19. Loshchilov, I., Hutter, F.: Decoupled weight decay regularization. In International Conference on Learning Representations (ICLR). (2019) https://doi.org/10.48550/arXiv.1711.05101
20. Luo, Z., Huang, Y., Liu, Q., Schwing, A. G.: Adaptive gradient methods with dynamic bound of learning rate. In International Conference on Learning Representations (ICLR). (2019) https://doi.org/10.48550/arXiv.1902.09843
21. Li, T., Zhou, P., He, Z., Cheng, X., Huang, X.: Friendly sharpness-aware minimization. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 5631-5640) (2024)
22. Mhmood, A.A., Ergül, Ö., Rahebi, J.: Detection of cyber-attacks on smart grids using improved VGG19 deep neural network architecture and Aquila optimizer algorithm. SIViP **18**, 1477–1491 (2024). https://doi.org/10.1007/s11760-023-02813-7
23. Nematollahi, M., Ghaffari, A., Mirzaei, A.: Task offloading in Internet of Things based on the improved multi-objective aquila optimizer. SIViP **18**, 545–552 (2024). https://doi.org/10.1007/s11760-023-02761-2
24. Nesterov, Y.: A method for solving the convex programming problem with convergence rate $O(1/k^2)$. Soviet Mathematics Doklady **27**, 372–376 (1983)
25. Reddi, S. J., Kale, S., Kumar, S.: On the convergence of Adam and beyond. In International Conference on Learning Representations (ICLR). (2019) https://doi.org/10.48550/arXiv.1904.09237
26. Robbins, H., Monro, S.: A stochastic approximation method. Ann. Math. Stat. **22**(3), 400–407 (1951). https://doi.org/10.1214/aoms/1177729586
27. Sivaprasad, S., et al.: On the Tunability of Optimizers in Deep Learning. (2019). arXiv preprint arXiv:1906.00807. https://doi.org/10.48550/arXiv.1906.00807
28. Sutskever, I., Martens, J., Dahl, G., Hinton, G.: On the importance of initialization and momentum in deep learning. In International Conference on Machine Learning (ICML), 1139–1147 (2013)
29. Sun, H., Cai, Y., Tao, R., Shao, Y., Xing, L., Zhang, C., Zhao, Q.: An Improved Reacceleration Optimization Algorithm Based on the Momentum Method for Image Recognition. Mathematics **12**(11), 1759 (2024). https://doi.org/10.3390/math12111759
30. Tieleman, T., Hinton, G.: Lecture 6.5–RMSprop: Divide the gradient by a running average of its recent magnitude. COURSERA Neural Networks for Machine Learning (2012)
31. Vaswani, A., et al.: Attention is all you need. In Advances in Neural Information Processing Systems (NeurIPS). (2017) https://doi.org/10.48550/arXiv.1706.03762
32. Wilson, A. C., et al.: The marginal value of adaptive gradient methods. In Advances in Neural Information Processing Systems (NeurIPS). (2017) https://doi.org/10.48550/arXiv.1705.08292
33. Zhang, C., Bengio, S., Hardt, M., Recht, B., Vinyals, O.: Understanding deep learning requires rethinking generalization. In International Conference on Learning Representations (ICLR). (2017) https://doi.org/10.48550/arXiv.1611.03530
34. Zhang, C., Recht, B.: Generalization in deep learning: Theory and practice. (2021) arXiv preprint arXiv:2102.01342. https://doi.org/10.48550/arXiv.2102.01342
35. Zuo, X., Zhang, P., Gao, S., Li, H. Y., Du, W. R.: NALA: a nesterov accelerated look-ahead optimizer for deep neural networks. (2023) https://doi.org/10.21203/rs.3.rs-3605836/v1
36. Zhou, P., Xie, X., Lin, Z., Yan, S.: Towards understanding convergence and generalization of AdamW. IEEE Trans. Pattern Anal. Mach. Intell. (2024). https://doi.org/10.1109/TPAMI.2024.3382294